(3) Move shaft by a distance of 12 inches and then place into the bearing. For this motion, the Analysis is *A* 12 *WSD* ; *W* is because of 7 lbs. of weight, *S*-due to steering of shaft end to the bearing and *D*-because shaft (or arm) will come to definite stop over the bearing.

Referring Table (*i*) *ARM*, the time units for a distance of 12 inches and three Work Factors, are 102. ...(*iii*)

Therefore total time units = (*i*) + (*ii*) + (*iii*)

= 85 + 38 + 102 = 225

Since 1 time unit = 0.0001 minutes, thus the total time for the existing work.

= 0.0001 × 225 = 0.0225 minutes

9.24. ACTIVITY SAMPLING, RATIO-DELAY STUDY OR WORK SAMPLING

Introduction. L.H.C. Tippett developed *Activity Sampling* in Britain in 1934 for the British Cotton Industry Research Board. R. L. Morrow used this technique in America round about 1945 and named it *Ratio Delay Study*. In 1952, C. L. Brisley renamed the technique as *Work Sampling* and today it is one of the very common techniques of Work Measurement.

Activity sampling as defined by B.S. 3138 : 1969, is a technique in which a large number of observations are made over a period of time of one or a group of machines, processes, or workers. Each observation records what is happening at that instant and the percent of observations recorded for a particular activity or delay is a measure of the percentage of time during which that activity or delay occurs.

Work Sampling can tell what percentage of the working day, a person spends how, *i.e.*, for how much time he works, what time he expends for his personal needs and for how long he remains idle.

Activities of very long duration (such as to find the actual working time of an operator in one shift of eight hours) cannot be economically timed with the help of stop watch time study. In such cases, the most appropriate technique of Work Measurement is Work Sampling which takes only 1/20th of the time required for stop watch study and gives the results within an accuracy of ± 2%.

Theory/Principle. Work sampling relies upon Statistical Theory of Sampling and Probability Theory. Normal Frequency Distribution and Confidence Level are associated very much with Work Sampling.

Statistical theory of sampling explains that adequate random samples of observations spread over a sufficient period of time can construct an accurate picture of the actual situation in the system. Approximately 500 observations produce fairly reliable results and the results obtained through observations 3000 or more are very accurate.

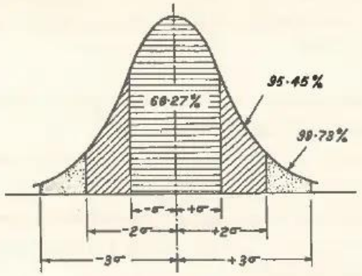

Fig. 9.17. Normal distribution curve and confidence level.

Normal frequency distribution (Fig. 9.17) portrays graphically the probability of occurrence of a chance event. Moreover, it gives important relationship between the number of standard deviations and the area under the curve representing the confidence level of an occurrence.

Standard deviation (σ)	Confidence level (%)
1	68.27
1.96	95.00
2	95.45
3	99.73

A confidence level of 95.45% signifies that the Work Study Engineer is sure that 95.45% of the times, the random observation will represent the true facts.

If the Work Study Engineer takes 25 rounds of the Machine Shop in a day, observes an operator 'x' and finds that

15 times he was working on the machine,

4 times he was setting up or cleaning the machine,

3 times he was not doing anything.

3 times he had gone for his personal needs ; it shows that the worker spends 60% of his time in actually working over the machines and for 12% of the total time he is idle, etc., etc. These facts can be confirmed or refuted by conducting more number of observations.

Procedure

(*a*) Define the problem, *i.e.*, determine the main objectives and define each activity to be measured.

(*b*) Make sure that all the persons connected with the study (*i.e.*, workers and supervisor) understand the objectives of the study.

(*c*) State the desired accuracy limits for the ultimate results.

(*d*) Conduct a Pilot Study to

(*i*) estimate the approximate percentage occurrence of the activity (*i.e.*, *p*).

(*ii*) estimate the required number of observations for the desired accuracy set, and

(*iii*) ensure that workers have become habituated to the visits of the work study engineer.

(*e*) Design the actual study.

(*f*) Using the data obtained from Pilot Study (*d*)(*i*), above, *i.e.*, value of *p*, calculate the number of observations to be made.

Example 9.3. Pilot Study showed the percentage of occurrence of an activity as 50%. Determine the number of observations for 95% confidence level and an accuracy of ±2%.

Solution

$$N = \frac{4p(100-p)}{L^2}$$

where N is the number of observations

p is % of occurence

L is limits of accuracy.

Therefore $N = \dfrac{4 \times 50(100-50)}{2^2} = 2500.$

(a) Other methods for finding number of observations are :

and Machine time $= 0.84 \times \frac{1}{3} = 0.28$ minutes

Normal time per article = Observed time×Rating+Machine time.

$= 0.56 \times 1.15 + 0.28$

$= 0.924$ minutes.

Standard time per article = Normal time+Allowances

$= 0.924 + 0.924 \times 12/100$

$= 1.0348$ minutes.

Advantages

(*i*) It involves much less cost as compared to stop watch time study.

(*ii*) It can be carried out with little training.

(*iii*) It can time long operations which are almost impractical to be measured (*i.e.*, timed) by stop watch time study.

(*iv*) It is very advantageous for timing group activities.

(*v*) It does not need any timing device like stop watch or microchronometer, etc.

(*vi*) Even if the study gets interrupted in between, it does not introduce any error in the results.

(*vii*) Observations can be made within the desired accuracy.

(*viii*) Large number of observations extended over days/weeks damp down the influence of day-to-day fluctuations on the results.

(*ix*) It can increase efficiency by uncovering the sources of delay.

Limitations

(*i*) It is uneconomical both as regards time and money to study activities of short duration by Work Sampling.

(*ii*) It is also uneconomical in case one worker or one machine is to be studied.

(*iii*) It does not break, the job into elements and thus does not provide element details.

(*iv*) It does not assist in improving work method.

(*v*) It normally does not account for the speed at which an operator is working.

(*vi*) Workers may not understand the principles of work sampling and hence may not trust it.

(*vii*) Observations, neither random nor sufficient in number may produce inaccurate results.

Applications/Uses

(*i*) To determine working time and idle time of men and machines.

(*ii*) To time long duration activities which are regular/irregular, frequent/infrequent.

(*iii*) To estimate the time for which material handling equipments are actually operating in a day.

(*iv*) To estimate allowances for unavoidable delay.

(*v*) In describing resource utilisation patterns.

(*vi*) For the purpose of cost control and accounting.

(*vii*) In estimating the percentage utility of the inspectors and time standards for indirect labour.

(*viii*) In stores, hospitals, warehousing, offices, farm work, repair and maintenance work, textile industry, machine shops, etc.

(*ix*) It is preferred when the cost of using other work measurement techniques for timing a job appears to be great.

9.25. SYNTHESIS OF WORK STUDY DATA

Work study data or in other words time study values of elements obtained from direct time studies, conducted on various jobs having different parameters and under different conditions are synthesized in order to obtain 'Synthetic Times' or 'Basic Data' or 'Standard Data'.

The 'Standard Data', its development and uses have been discussed in detail in Section 9.19.

Before describing the procedure for synthesis of work study data, it is better to understand the meaning of synthesis. *Synthesis* is a work measurement technique to build up normal time for a new job (at a defined level of performance) by adding element times collected from previously held time studies on similar jobs having same elements as possessed by the new job.

The following steps are involved in synthesis :

(1) Collect all the possible details about the job, for example, material, dimensions, method, conditions, etc. The collected details should be reliable.

(2) Break the job into constituent elements. The size of the elements should be decided critically. A smaller element will have much wider range of usefulness and applicability but the time study Engineer will naturally take longer time to compile the standard time for the job.

Three types of elements, namely machine elements, constant elements, and variable elements may be there, in a job.

Machine Elements. They are controlled by the characteristics of the process such as feed, speed, depth of the cut, amount of metal to be removed, etc.

Constant Elements. They are identical from job to job; for example, mounting a cutter on the milling machine arbor or holding a lathe tool in the tool post.

Variable Elements. They are similar in nature from job to job but vary in difficulty and the time required to complete them, because different jobs may possess different dimensions, shapes, and weights. Variable elements may be those, involving hand work or others, influenced by the changes in metal machinability, quality, tolerances, etc.

Constant, variable or machine elements in each job, may be repeated the same number of time, different number of times or they may or may not be present in various jobs.

Constant elements are easy to deal with. Only sufficient number of observations are required to obtain a realistic time. Variable elements are quite problemous as compared to constant elements and need more skill and attention on the part of analyst. Good number of observations are must in order to establish a relationship (which may be a straight line or a curve) between the element characteristics and basic time for that element.

Machine elements are normally calculated from the information, of speed of job rotation, feed of the tool and the depth of cut, etc.

(3) As far as possible select the appropriate normal times for all the elements involved in the operation, from the synthetic data or the standard data.

(4) Estimate various allowances like, personal and rest allowances, process allowances and special allowances, for each element.

(5) Verify the analysis of elements for the selected job method and other conditions.

(6) Add various allowances to the normal time for each element and sum up all such times to fix the standard time for the new job.

9.26. USE OF TIME STUDY DATA IN WAGE INCENTIVE AND COLLECTIVE BARGAINING

This topic involves three meaningful words which need some elaboration. They are :

(*a*) *Time Study Data.* It contains the facts or the informations resulted from time study. Time values

Analytical Estimating differs from synthesis in the sense that it estimates time of non-repetitive and long operation jobs, for whose all job elements, past time data may not be available with the firm. Hence the time for the new job has to be determined, by timing some elements on the basis of synthetic data available and the remaining elements are given a time by the estimator from his past experience of timing different jobs.

9.21. PREDETERMINED MOTION TIME SYSTEM (PMTS) OR ELEMENTAL MOTION TIME SYSTEM OR BASIC MOTION TIME SYSTEM

Introduction. In section 9.19, the Standard Data has been classified as *Macro Data and Micro Data.* An element in Macro data, *e.g.,* pick up the screw driver may have its timed value of several seconds whereas in Micro data the elements of the job are basic human motions wth duration 0.1 second or even less. This section will deal with *M-T-M and Work Factor Systems (i.e.,* PMTS) which are known as Micro data. Micro data is based upon much smaller division of motions (*i.e.,* therbligs) as compared to Macro data.

BS 3138 : 1969 *defines* PMTS as a work measurement technique whereby times established for basic human motions (classified according to the nature of the motion and the conditions under which it is made) are used to build up the time for a job at a defined level of performance.

Technique :

(*i*) The technique to build PMTS data does not measure element time by a stop watch and thus it avoids the inaccuracies being introduced owing to the element of human judgement.

(*ii*) It is assumed that all manual tasks in industries are made up of certain basic human movements (like reach, move, disengage etc.) which are common to almost all jobs.

(*iii*) The average time taken by the (normal) industrial workers to perform a basic movement is practically constant.

With the above facts in mind, the various steps involved in collecting PMTS data are as follows :

(*a*) Select large number of workers doing varieties of jobs under normal working condtions in industries.

(*b*) Record the job operations on a movie film. (Micromotion study).

(*c*) Analyse the film, note down the time taken to complete each element and compile the data in the form of a table or chart.

The jobs selected are such that they involve most of the common basic motions and are worked under different set of conditions by workers having different ages and other characteristics.

Once the tables for various basic motions are ready, the normal time for any new job can be determined by breaking the job into its basic movements, noting time for each motion from the tables and adding up the time values for all the basic motions involved in the job.

Standard time may be obtained by adding proper allowances.

Objects and Uses of PMTS. Predetermined motion time system finds the following uses :

(*i*) It is very useful in Method Analysis.

(*ii*) It helps modifying and improving work methods before starting the work on the job.

(*iii*) It sets time standards for different jobs.

(*iv*) It assists in constructing time formulae.

(*v*) It aids in the prebalancing of the manufacturing lines.

(*vi*) It provides a basis for wage plans and labour cost estimation.

(*vii*) It facilitates training of the workers and supervisor.

(*viii*) It s used for timing those short and repetitive motions which cannot be measured by stop watch.

Advantages of PMTS. Predetermined motion time system possesses the following advantages :

(*i*) It eliminates inaccuracies associated with stop watch time study.

(*ii*) It is superior to stop watch time study when applied to short cycle highly repetitive operations.

(*iii*) Time standard for a job can be arrived at without going to the place of work.

(*iv*) Unlike stop watch study, no rating factor is employed.

(*v*) PMTS data, since it is the result of very large number of observations, is more reliable and accurate as compared to stop watch time study data.

(*vi*) The time and cost associated with finding the standard time for a job is considerably reduced.

(*vii*) Alternative methods are compared easily.

(*viii*) PMTS helps in tool and product design.

Drawbacks :

(*i*) PMTS can deal only with manual motions of an operation.

(*ii*) All categories of motions have not been considered while collecting PMTS data.

Applications of PMTS. Predetermined motion time system is usefully employed for the following types of jobs :

(*a*) Machining work,

(*b*) Maintenance work,

(*c*) Assembly jobs,

(*d*) Servicing, and

(*e*) Office work.

9.22. METHOD-TIME-MEASUREMENT (M-T-M) :

Introduction. This system was developed by Method-Time-Measurement Association and it got recognition in 1948. Unlike work factor system, M-T-M does not require any modification of the basic time values. Moreover the basic human movements in this system are analysed in more detail. M-T-M measures time in terms of TMUs (Time-Measurement-Units) and 1 TMU=0.0006 minutes.

M-T-M analyses an industrial job into the basic human movements required to do the same. From the tables of these basic motions (Table I to IX), depending upon the kind of motion, and conditions under which it is made, predetermined time values are given to each motion. When all such times are added up, it provides the normal time for the job. Standard time can be found by adding suitable allowances.

According to M-T-M the various classification of motions are :

(*i*) Reach-R,

(*ii*) Move-M,

(*iii*) Turn and Apply Pressure-T and AP,

(*iv*) Grasp-G,

(*v*) Position-P,

(*vi*) Release-RL,

(*vii*) Disengage-D,

(*viii*) Eye travel time and eye focus time-ET and EF.

(*ix*) Body, Leg and Foot Motions, and

(*x*) Simultaneous Motions.